

# Support Vector Machine for Non-Linearly Separable Patterns

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INDRAPRASTHA INSTITUTE *of*  
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DELHI



# Non-linearly Separable Data

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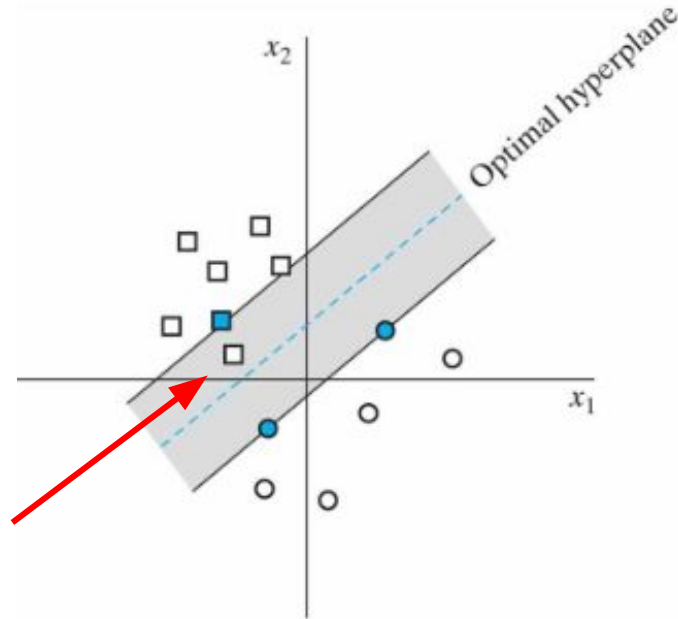
- Linear SVM with Soft Margin
- Non-linear SVM
  - Kernel SVM



# Non-Separable Patterns



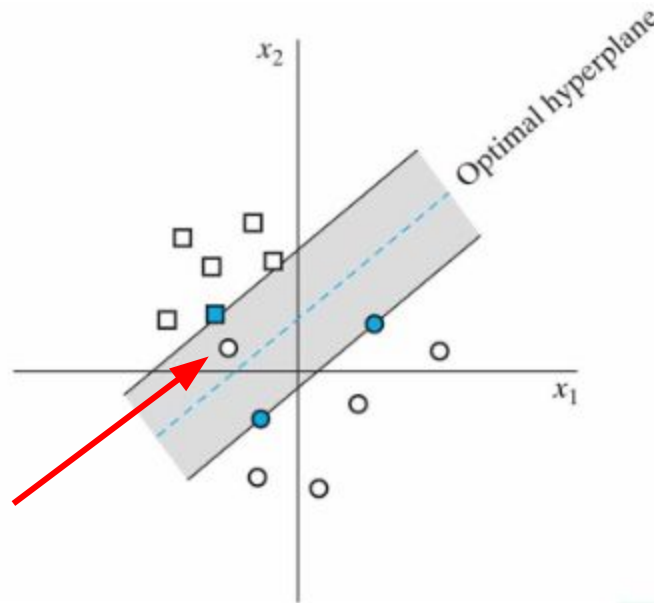
- $y_i^*(\mathbf{w}^T \mathbf{x}_i + b) \geq 1, \forall i=1,2, \dots, N$
- *Violation:*
  - $\{(x_i, y_i)\}$  falls on the right side of the decision surface



# Non-Separable Patterns



- $d_i^*(\mathbf{w}^T \mathbf{x}_i + b) \geq 1, \forall i=1,2, \dots, N$
- *Violation:*
  - $\{(x_i, y_i)\}$  falls on the wrong side of the decision surface



# Quiz

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<https://forms.gle/RghtRV1KEbMNbfPF9>



# Slack Variables



- Let  $\{\epsilon_i\}_{i=1}^N \geq 0$ 
  - $y_i^*(\mathbf{w}^T \mathbf{x}_i + b) \geq 1 - \epsilon_i, \forall i=1,2, \dots, N$
- $\epsilon_i$  are known as slack variables:
  - Deviation of a data point from the ideal condition of pattern separability
    - $0 \leq \epsilon_i \leq 1$ : Data point falls on the right side of the decision surface
    - $\epsilon_i > 1$ : Data point falls on the wrong side of the decision surface
- Support vectors are those particular data points that satisfy the equation precisely even if  $\epsilon_i > 0$

# Optimal Hyperplane



- The cost function
  - $\Phi(\mathbf{w}, \xi) = \frac{1}{2} \mathbf{w}^T \mathbf{w} + C \sum_{i=1}^N \xi_i$
  - C: Regularization parameter that controls the tradeoff between the complexity of the machine and the number of nonseparable patterns.
- High value of C -> High confidence in the quality of the training data
- Small value of C -> Less emphasis on the training data
- C has to be selected by the user *experimentally*.

# Primal Problem



- Given the training sample  $\{(\mathbf{x}_i, y_i)\}_{i=1}^N$ , find the optimum values of the weight vector  $\mathbf{w}$  and bias  $b$  such that they satisfy the constraints
  - $y_i^*(\mathbf{w}^T \mathbf{x}_i + b) \geq 1 - \varepsilon_i, \forall i=1,2, \dots, N$
  - $\varepsilon_i \geq 0, \forall i$
- and such that the weight vector  $\mathbf{w}$  and the slack variables  $\varepsilon_i$  minimize the cost functional
  - $\Phi(\mathbf{w}, \varepsilon) = 1/2 \mathbf{w}^T \mathbf{w} + C \sum_{i=1}^N \varepsilon_i$where  $C$  is user-specified positive parameter.
- *Exercise: Apply the Lagrangian multipliers to get the dual*



# Dual Problem

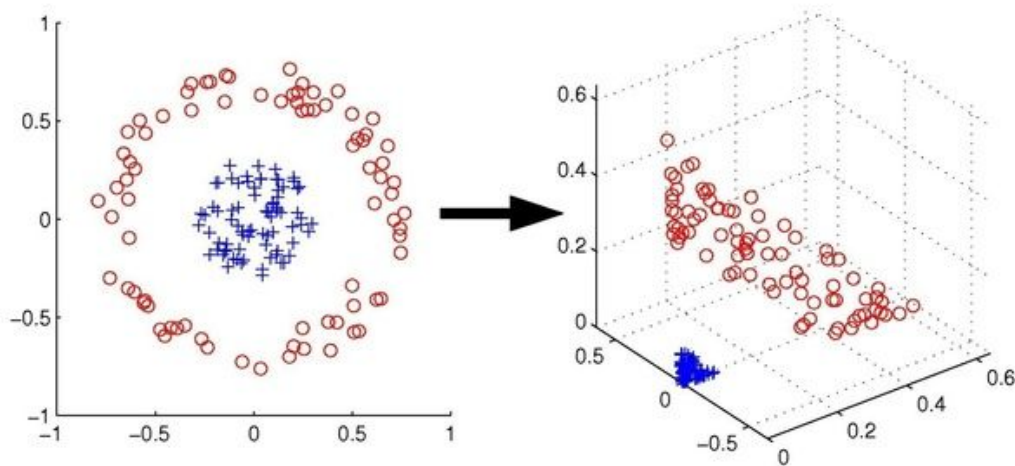


- Given the training sample  $\{(\mathbf{x}_i, y_i)\}_{i=1}^N$ , find the Lagrange multipliers  $\alpha_i$  that maximize the objective function
  - $Q(\alpha) = \sum_{i=1}^N \alpha_i - 1/2 \sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j y_i y_j \mathbf{x}_i^T \mathbf{x}_j$ , S.T.
    - $\sum_{i=1}^N \alpha_i y_i = 0$
    - $0 \leq \alpha_i \leq C, \forall i = 1, 2, \dots, N$where  $C$  is user-specified positive parameter
- Neither the slack variables nor their own Lagrange multipliers appear in the dual problem!
- Separable vs Non Separable
  - $\alpha_i \geq 0$
  - $0 \leq \alpha_i \leq C, \forall i = 1, 2, \dots, N$

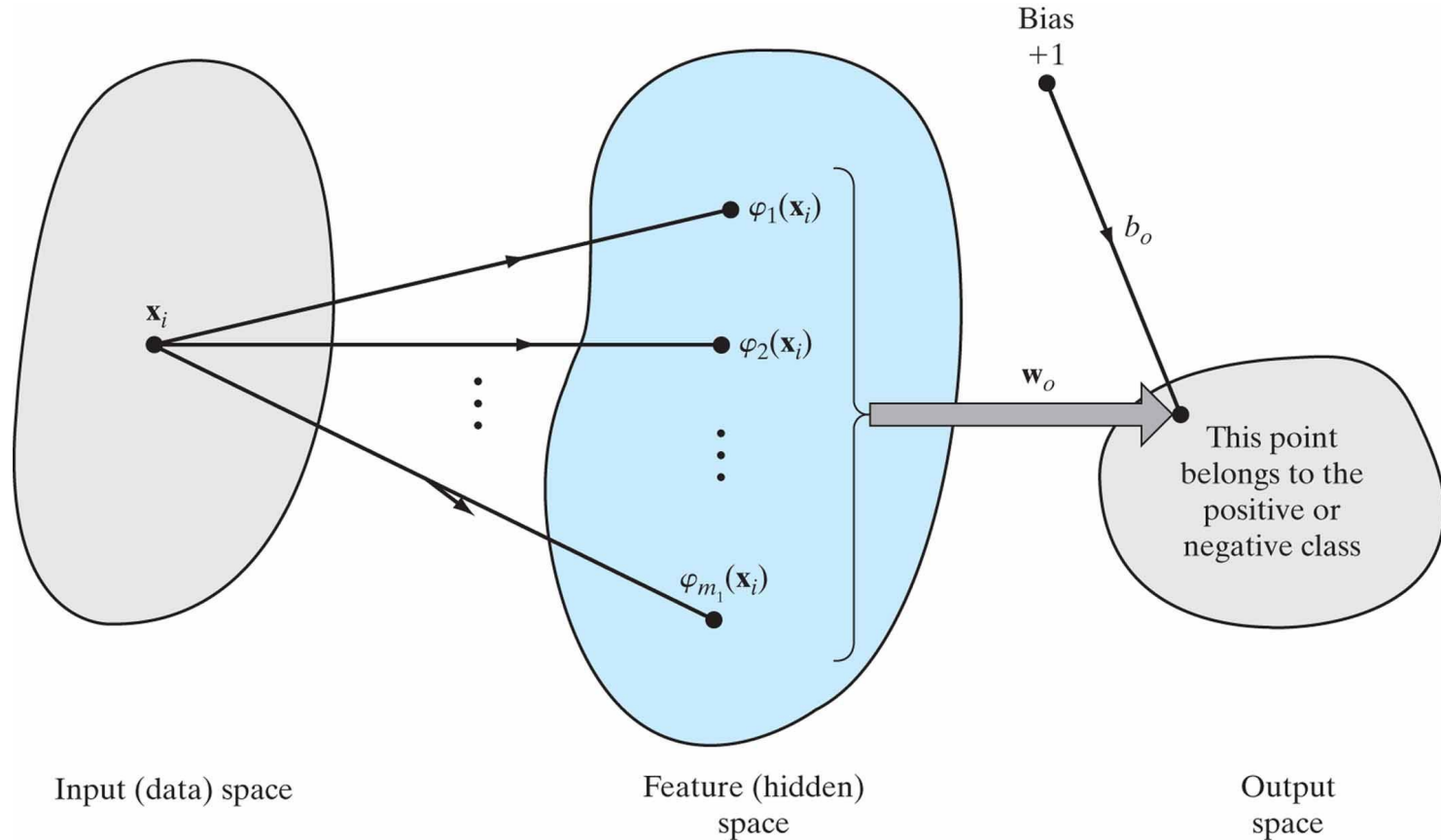
# Cover's Theorem:(Cover, 1965)



- “A complex pattern-classification problem cast in high-dimensional space nonlinearly is more likely to be linearly separable than in a low dimensional space”
  - Transformation is non-linear
  - Dimensionality of the feature space is high enough



# Kernel SVM: Non-linear Mapping



# Kernel SVM: Optimal Hyperplane in Feature Space



- Similar to the idea of the non separable patterns
  - Separating hyperplane is defined as a linear function of vectors drawn from the feature space rather than the original input space.

- Non-linear Transformation

$$\Phi(x) = [\Phi_0(x), \Phi_1(x), \dots, \Phi_m(x)]$$

$$\mathbf{w} = \sum_{i=1}^N \alpha_i y_i x_i \quad \mathbf{w} = \sum_{i=1}^N \alpha_i y_i \Phi(x_i)$$

$$\mathbf{w}^T \Phi(x) = 0 \quad \sum_{i=1}^N \alpha_i y_i \Phi^T(x_i) \Phi(x) = 0$$

# The Inner Product Kernel



- For a given set of training samples (in a lower-dimensional feature space) and a transformation into a higher-dimensional space, there exists a function (The Kernel Function) which can compute the dot product in the higher-dimensional space without explicitly transforming the vectors into the higher-dimensional space first.

$$\begin{aligned} K(\mathbf{x}, \mathbf{x}_i) &= \Phi(\mathbf{x})^T \Phi(\mathbf{x}_i) \\ &= \sum_{j=0}^m \Phi_j(x)^T \Phi_j(x_i), \forall i = 1, 2, \dots, N \\ K(\mathbf{x}, \mathbf{x}_i) &= K(\mathbf{x}_i, \mathbf{x}), \forall i \end{aligned}$$

# Quiz

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<https://forms.gle/1Wh82RYiG9tr5qAs6>



Is the Kernel Matrix  $K$  symmetric?

- A. Yes, since  $k(x, y) = k(y, x) = \phi(x) \cdot \phi(y)$
- B. No, since  $K$  will not be a square matrix
- C. No,  $k(x, y) \neq k(y, x)$
- D. Can't say

# The Kernel Trick



- Optimal Hyperplane

$$\sum_{i=1}^N \alpha_i y_i K(\mathbf{x}, \mathbf{x}_i) = 0$$

- Specifying the kernel is *sufficient* for the pattern classification in the output space; we need never explicitly compute the weight vector  $\mathbf{w}_0$
- The optimal hyperplane consists of a finite number of terms that is equal to the number of support vector patterns.



# Optimum Design of a SVM



- Given the training sample  $\{(\mathbf{x}_i, y_i)\}_{i=1}^N$ , find the Lagrange multipliers  $\alpha_i$  that maximize the objective function
  - $Q(\alpha) = \sum_{i=1}^N \alpha_i - 1/2 \sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j y_i y_j K(\mathbf{x}_i, \mathbf{x}_j)$  S.T.
    - $\sum_{i=1}^N \alpha_i y_i = 0$
    - $0 \leq \alpha_i \leq C, \forall i = 1, 2, \dots, N$where  $C$  is user-specified positive parameter
- $K(\mathbf{x}_i, \mathbf{x}_j)$  can also be viewed as the  $ij$ -th element of a symmetric  $N$ -by- $N$  matrix  $K$ :
  - $K = \{K(\mathbf{x}_i, \mathbf{x}_j)\}_{(i,j) \in \bar{N}_s \times \bar{N}_s}^N$
- Optimal weight  $\mathbf{w}_o = \sum_{i=1}^{\bar{N}_s} \alpha_{o,i} y_i \boldsymbol{\varphi}(\mathbf{x}_i)$ 
  - First component of  $\mathbf{w}_o$  represents the bias  $b_o$
- Feature Space dimensionality is determined by  $\bar{N}_s$ .

# Inner-Product Kernels



TABLE 6.1 Summary of Mercer Kernels

| Type of support vector machine | Mercer kernel<br>$k(\mathbf{x}, \mathbf{x}_i), i = 1, 2, \dots, N$      | Comments                                                                                   |
|--------------------------------|-------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Polynomial learning machine    | $(\mathbf{x}^T \mathbf{x}_i + 1)^p$                                     | Power $p$ is specified <i>a priori</i> by the user                                         |
| Radial-basis-function network  | $\exp\left(-\frac{1}{2\sigma^2} \ \mathbf{x} - \mathbf{x}_i\ ^2\right)$ | The width $\sigma^2$ , common to all the kernels, is specified <i>a priori</i> by the user |
| Two-layer perceptron           | $\tanh(\beta_0 \mathbf{x}^T \mathbf{x}_i + \beta_1)$                    | Mercer's theorem is satisfied only for some values of $\beta_0$ and $\beta_1$              |

- *Exercise: Apply the Lagrangian multipliers to get the dual*
- *Exercise: Read Mercer Theorem*

# Quiz

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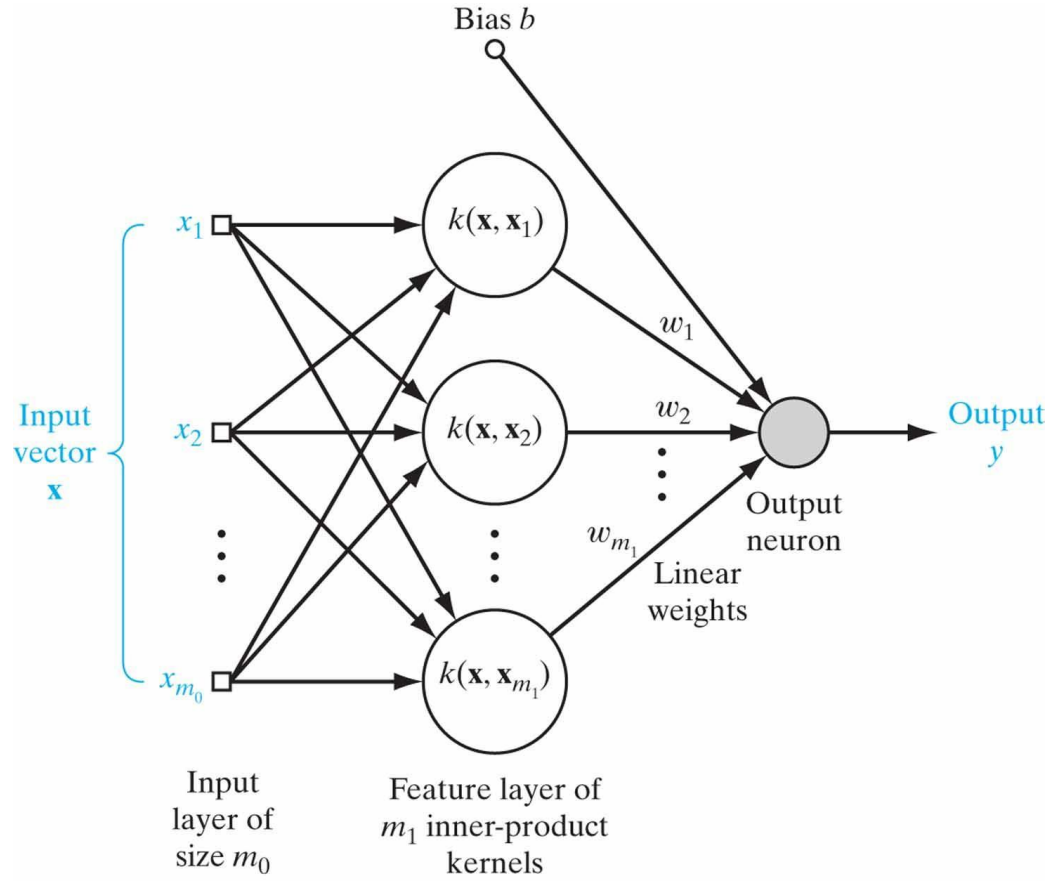


The number of arithmetic operations required in the kernel polynomial function  $(1 + \mathbf{x}_i^T \mathbf{x}_j)$  is more than the dot product?

- A. Yes
- B. No
- C. Not defined



# Architecture of SVM

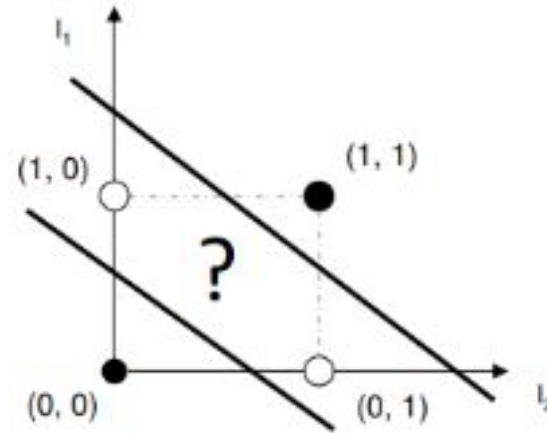


# XOR Problem: Kernel Trick



TABLE 6.2 XOR Problem

| Input vector $\mathbf{x}$ | Desired response $d$ |
|---------------------------|----------------------|
| $(-1, -1)$                | -1                   |
| $(-1, +1)$                | +1                   |
| $(+1, -1)$                | +1                   |
| $(+1, +1)$                | -1                   |



- Given the following kernel:

- $K(\mathbf{x}_i, \mathbf{x}_j) = (1 + \mathbf{x}_i^T \mathbf{x}_j)^2$

Derive the optimum  $\mathbf{w}$  for a SVM using the above kernel

# Example: XOR Problem



- Define a kernel:
  - $K(\mathbf{x}_i, \mathbf{x}_j) = (\mathbf{1} + \mathbf{x}_i^T \mathbf{x}_j)^2$
- Let  $\mathbf{x} = [x_1, x_2]^T$ ,  $\mathbf{x}_i = [x_{i1}, x_{i2}]^T$
- $K(\mathbf{x}_i, \mathbf{x}_j) = ?$ 
  - $1 + x_1^2 x_{i1}^2 + 2x_1 x_2 x_{i1} x_{i2} + x_2^2 x_{i2}^2 + 2x_1 x_{i1} + 2x_2 x_{i2}$
- $\varphi(\mathbf{x}) = [1, x_1^2, \sqrt{2}x_1 x_2, x_2^2, \sqrt{2}x_1, \sqrt{2}x_2]$
- $\varphi(\mathbf{x}_i) = [1, x_{i1}^2, \sqrt{2}x_{i1} x_{i2}, x_{i2}^2, \sqrt{2}x_{i1}, \sqrt{2}x_{i2}]$
- Gram

$$\mathbf{K} = \begin{bmatrix} 9 & 1 & 1 & 1 \\ 1 & 9 & 1 & 1 \\ 1 & 1 & 9 & 1 \\ 1 & 1 & 1 & 9 \end{bmatrix}$$

# Example: XOR Problem



- Dual form Objective Function

- $Q(\alpha) = \sum_{i=1}^N \alpha_i - 1/2 \sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j y_i y_j K(\mathbf{x}_i, \mathbf{x}_j),$

$$Q(\alpha) = \alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 - \frac{1}{2} (9\alpha_1^2 - 2\alpha_1\alpha_2 - 2\alpha_1\alpha_3 + 2\alpha_1\alpha_4 \\ + 9\alpha_2^2 + 2\alpha_2\alpha_3 - 2\alpha_2\alpha_4 + 9\alpha_3^2 - 2\alpha_3\alpha_4 + 9\alpha_4^2)$$

- *Conditions of optimality*

- $\partial Q(\alpha)/\partial \alpha = 0$

$$\begin{aligned} 9\alpha_1 - \alpha_2 - \alpha_3 + \alpha_4 &= 1 \\ -\alpha_1 + 9\alpha_2 + \alpha_3 - \alpha_4 &= 1 \\ -\alpha_1 + \alpha_2 + 9\alpha_3 - \alpha_4 &= 1 \\ \alpha_1 - \alpha_2 - \alpha_3 + 9\alpha_4 &= 1 \end{aligned}$$



# Example: XOR Problem



- Dual form Objective Function Solution

$$\alpha_{o,1} = \alpha_{o,2} = \alpha_{o,3} = \alpha_{o,4} = \frac{1}{8}$$

- Optimum Weight

- $\mathbf{w}_o = \sum_{i=1}^{N_s} \alpha_{o,i} d_i \varphi(\mathbf{x}_i)$

$$\mathbf{w}_o = \frac{1}{8} [-\varphi(\mathbf{x}_1) + \varphi(\mathbf{x}_2) + \varphi(\mathbf{x}_3) - \varphi(\mathbf{x}_4)]$$

# Example: XOR Problem



- Optimum Weight

$$\frac{1}{8} \left[ - \begin{bmatrix} 1 \\ 1 \\ \sqrt{2} \\ 1 \\ -\sqrt{2} \\ -\sqrt{2} \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ -\sqrt{2} \\ 1 \\ -\sqrt{2} \\ \sqrt{2} \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ -\sqrt{2} \\ 1 \\ \sqrt{2} \\ -\sqrt{2} \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \\ 1 \\ \sqrt{2} \\ \sqrt{2} \\ \sqrt{2} \end{bmatrix} \right]$$

$$\begin{bmatrix} 0 \\ 0 \\ -1/\sqrt{2} \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

# Example: XOR Problem



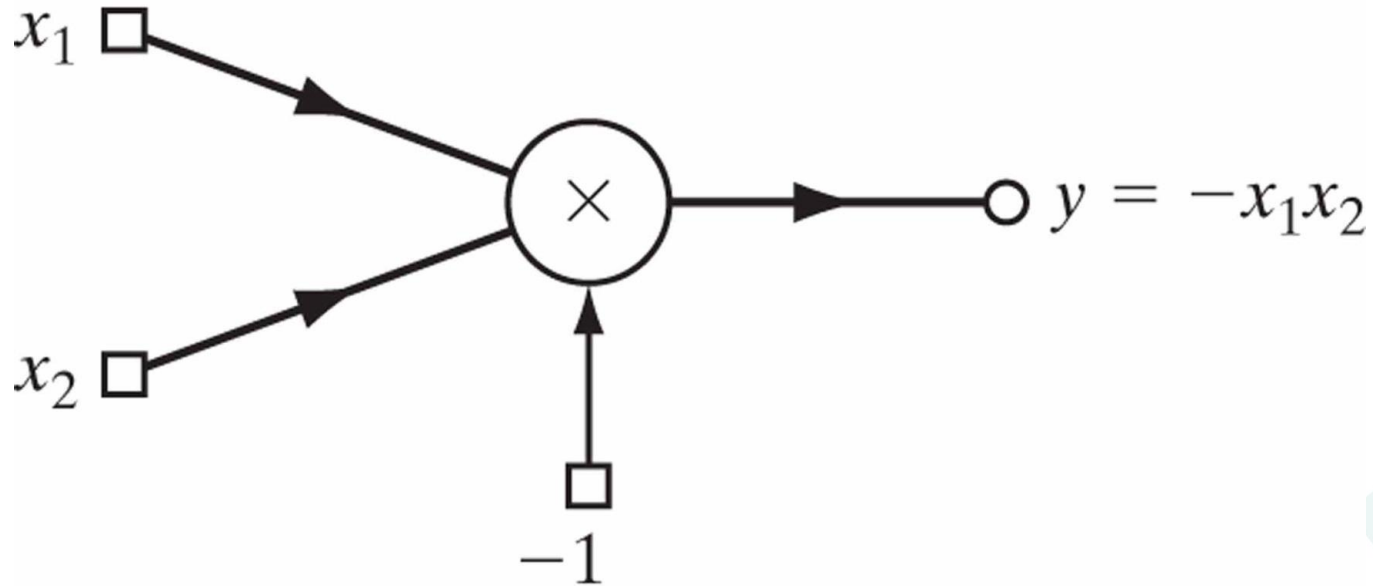
- Optimum Hyperplane
  - $\mathbf{w}_o^T \boldsymbol{\varphi}(\mathbf{x}) = 0$  or

$$\begin{bmatrix} 0, 0, \frac{-1}{\sqrt{2}}, 0, 0, 0 \end{bmatrix} \begin{bmatrix} 1 \\ x_1^2 \\ \sqrt{2}x_1x_2 \\ x_2^2 \\ \sqrt{2}x_1 \\ \sqrt{2}x_2 \end{bmatrix} = 0$$

$$\sum_{i=1}^N \alpha_i d_i K(\mathbf{x}, \mathbf{x}_i) = 0$$

$$-x_1x_2 = 0$$

# Example: XOR Problem



(a)

# References



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